Hantian Zhang

hantianzhang0@gmail.com

Education

Jinan University **Guangzhou, China**
B.Sc. in Information and Computing Science, Department of Mathematics *Sep. 2023 – Present*

- GPA: 3.83/5.0 (88.3/100) | IELTS: 7.0 | CET-6: 587

Research Interest

AI Security, Trustworthy AI, Diffusion Models

Publications and Manuscripts

- **Hantian Zhang**, Chenru Wang, Xinyi Ye, Xiaoxi Wei, & Chi Zhang (2026). GTS: Gap-targeted synthesis for budget-efficient synthetic data scaling [Submitted to **ICLR 2027**].
- **Hantian Zhang** & Wentai Wu (2026). Do LLMs Know When Evidence is Insufficient? An Evidence Sufficiency Benchmark for Answer-Abstention Calibration in Retrieval-Augmented Generation. ***Computers, Materials & Continua***, 89(1). doi:10.32604/cmc.2026.086343.
- **Hantian Zhang**, Zhijun Ji, & Wentai Wu (2026). Utility-Constrained Adaptive Patch Perturbation for Privacy-Preserving Face-Image Retrieval [Accepted]. ***Computers, Materials & Continua***.
- **Hantian Zhang**, Linghang Zeng, & Wentai Wu (2026). RAGPos-Bench: Diagnosing Evidence Position Bias in Retrieval-Augmented Generation [Accepted regular paper]. ***Pacific Rim International Conference on Artificial Intelligence*** (PRICAI 2026).

Research Experience

Westlake University, AGI Lab **Hangzhou, China**
Visiting Student; Advisors: Asst. Prof. Chi Zhang and Chenru Wang *Mar. 2026 – Present*

- Proposed Gap-Targeted Synthesis (GTS), a budget-efficient data-scaling method that targets what is missing from real training data instead of adding randomly generated samples.
- Developed a sampling-time guidance method that steers diffusion models toward distributional gaps identified directly from real data.

Nankai University **Tianjin, China**
Visiting Student, Advisor: Assoc. Prof. Qingkai Zeng *Jan. 2026 – Present*

- Designed a benchmark framework for evaluating stance bias in social event search results across commercial search engines (Google, Bing, Baidu).
- Built a multi-level annotation taxonomy classifying search-result bias along political, ideological, and framing dimensions.
- Investigated evidence reliability and conflict resolution strategies in RAG systems under incomplete information.
- Proposed fairness evaluation protocols for LLM-based retrieval systems, focusing on robustness under adversarial and conflicting evidence.

Jinan University **Guangzhou, China**
Research Assistant, Advisor: Assoc. Prof. Wentai Wu *Nov. 2025 – Present*

- Proposed a dual-score adaptive patch perturbation method for privacy-preserving face-image retrieval, combining reconstruction sensitivity and CLIP-based semantic privacy relevance while preserving retrieval utility; evaluated on FFHQ, LFW, and AgeDB with 300 queries per dataset.
- Developed RAGPos-Bench, a controlled benchmark that varies evidence order while holding content fixed to isolate position bias and conflict handling in RAG; evaluated seven LLMs on 15,000 instances, revealing conflict-order sensitivity even in high-accuracy models.

Teaching Experience

Teaching Assistant **Jinan University**
Calculus, Linear Algebra *Mar. 2025 – Present*

- Taught Calculus and Linear Algebra in review sessions and helped students solve course problems.

Competition Experience and Honors

- First-Class Outstanding Student Scholarship, Jinan University, 2026–2027
- Third-Class Outstanding Student Scholarship, Jinan University, 2025–2026
- Outstanding Student Leader Scholarship, Jinan University, 2024–2025
- Outstanding Student Leader (Honorary Title), Jinan University, 2026–2027
- "Five-A Outstanding Student," Jinan University, 2025
- Selected Participant, International Elite Talent Training Program (Certified by China International Education Institute), 2024
- Second Prize, Asia-Pacific Mathematical Contest in Modeling (APMCM), 2025
- Third Prize, Asia-Pacific Mathematical Contest in Modeling (APMCM), 2024
- Second Prize, MathorCup Mathematical Modeling Competition, 2025

Skills

- **Programming:** Python, C, MATLAB, SQL
- **Frameworks and Tools:** PyTorch, Transformers, HuggingFace, vLLM, Git, Linux, LaTeX, Neo4j
- **Languages:** Chinese (Native); English (CET-6: 587, IELTS: 7.0)